

U.S. Beef Traceability Project Report

Sustainable Business Development
December 2023

Rabobank

Introduction

This report is an overview of Rabobank's project working with multiple partners to trace the sustainability footprint of beef cattle from birth to harvest through a blockchain traceability concept.

Rabobank's mission is Growing a Better World Together, and the Bank is committed to financing the Food System Transition so that the global food and agricultural system can sustainably feed 10 billion people by 2050.

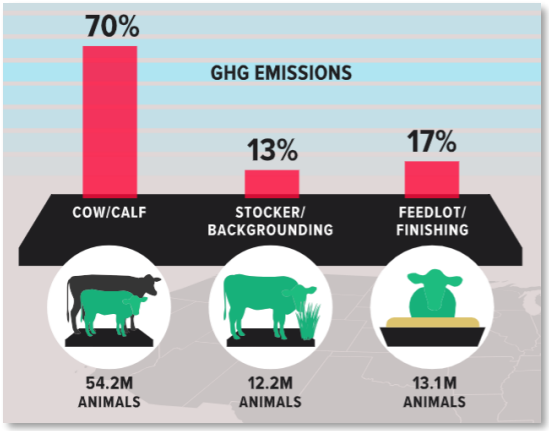
The US beef sector is one of the most important agricultural industries in the United States, consistently accounting for one of the largest shares of total cash receipts for agricultural commodities (USDA ERS, 2023). At the same time, the beef sector has one of the highest GHG footprints per unit of output, globally and in the US. Through Rabobank's commitment to the UN Net Zero Banking Alliance (UNZBA), the Bank calculates its own financed emissions (Scope 1, 2, and 3) and US beef emerges as a major hotspot due to the sector's high carbon footprint and the Bank's significant US beef portfolio (Rabobank, 2022). Even though US is one of the most efficient beef producers globally, significant efforts are still needed to reduce the sector's climate and environmental impact.

Rabobank is uniquely positioned to support its clients through the transition to more sustainable beef given our important role as lender-connector-influencer throughout the value chain. Simply divesting, as some stakeholders might advocate, does not equate to a climate transition.

Notice: This document is informational in nature, contains sensitive information, and is intended for the recipient only. Please contact Michael Rinaldi, Head of Sustainable Business Development, Rabobank North America Michael.Rinaldi@Rabobank.com with any questions.

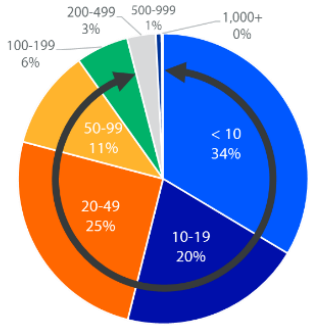
The Challenge

Most of the climate impact, around 70%, from the US beef industry is allocated to the highly dispersed cow-calf stage of the supply chain where cattle spend most of their life, primarily on pasture. This is due to the high number of animals residing in the segment for a longer period of time, generally lower-quality diet, and feed conversion efficiency (Beef Checkoff, 2020). The average beef cattle herd size is 44 head, with 93% of operations housing less than 200 animals and most relying on other sources of income including row crops or off-farm employment (USDA, Rabobank 2023). Additionally, cattle often move throughout state-lines, having multiple stops before reaching the processor (cow-calf ranch, auction market, backgrounder, feedyard) and creating a logistical challenge for emissions allocation and tracking.



Source: Beef Checkoff, 2020

2017 U.S. Beef Cow Operations by Herd Size



The last Census of Agriculture found 729, 000 beef cow operations, around 93% of operations have <200 cows.

Source: USDA, Rabobank 2023

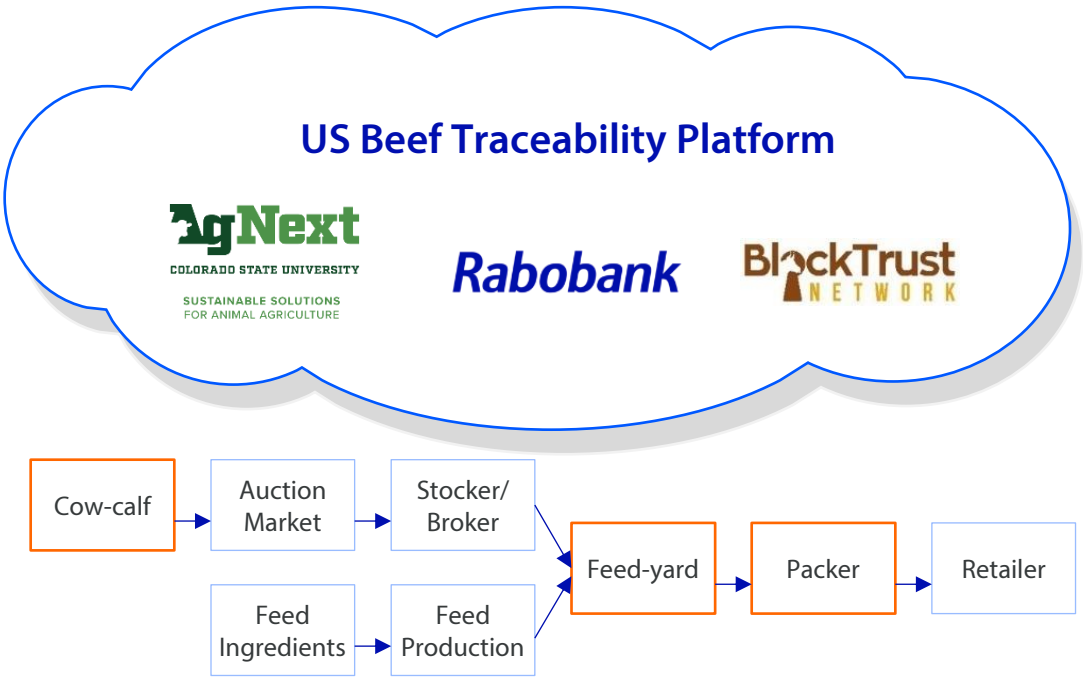
Beef processors receive a majority of the industry's scrutiny due to the phase being highly concentrated on four major players who control about 85% of the market: Tyson, Cargill, JBS, and National Beef (Reuters, 2021). These leading US beef processors have ambitions to reduce the climate impacts of their businesses, will all four committing to Science Based Targets (SBTi). To demonstrate progress against climate goals, beef companies need to track and verify the sustainability of beef production at each step of the supply chain to create an accurate, aggregated, and transparent sustainability view. Companies are also facing pressure to have clear evidence backing sustainable labelling or product claims to gain additional value for emission reduction. While the cow-calf sector poses the greatest logistical challenge to supply chain traceability, it also serves as a major opportunity to drive progress in sustainability.

In addition to voluntary climate action, regulation related to animal welfare, biosecurity, emissions reporting, and trade may also accelerate the need for strong traceability frameworks. Current traceability requirements in the US include the "Traceability for Livestock Moving Interstate" rule requiring cattle be officially identified and documented during certain interstate movements to prevent disease spread, implemented in 2013 by the USDA (USDA APHIS, 2023).

In the US, there has been a strong culture of independence in the beef industry with little data sharing, unless legally required, occurring between each stage of the supply chain (GreenBiz, 2014). In most cases, cow calf ranchers have no idea where their cattle ends up post auction while beef packers have no idea what farms their cattle originated from. This data disconnect means that cow calf producers have no incentive to optimize production beyond weight, and beef packers can't accurately measure the sustainability of their supply chain. While there is a limited pool of intervention options available to cow-calf producers (including feed additives, genetic selection, and production practices), cost, scalability, and feasibility remain the largest roadblocks to adoption.

Project Overview

Rabobank launched a pilot in 2021 to address this data challenge, utilizing blockchain technology for traceability and the U.S. Roundtable for Sustainable Beef's (USRSB) self-reported sustainability assessment for measuring sustainability performance of the supply chain players (USRSB Self-Assessment, 2023). The goal of the pilot was to provide a proof of concept, identify infrastructure, data, financial needs, and create a business case and viable business model for sustainability focused traceability in the beef industry.



Initially, the US Beef Traceability Project focused on tracing cattle from Cow-calf ranches, to the Feed-yard, and the Packer.

Core partners:

- BlockTrust Network – a transparency platform designed to connect partners across the beef supply chain. BlockTrust Network uses APIs to share carcass data, animal metrics, and animal-level information to the customer.
- Colorado State University's AgNext – a leading research collaborative for animal and ecosystem health. The collaborative works with producers, industry partners, and researchers to identify and innovate solutions for sustainability in animal agriculture.
- MFA – a Midwest-based regional farm supply and marketing cooperative. MFA is a primary manufacturer of livestock feed, as well as a supplier and marketer of plant food and crop protection products. MFA currently has 45,000 members in Missouri and adjacent states.
- Feedlots – industry leading feedlots committed to the economic, environmental, and social pillars of sustainability
- Packers – major beef processors representing significant US market share have ambitious sustainability and climate ambitions, seeking to demonstrate progress through traceability

The project initially focused on the beginning of the supply chain, working with the Missouri based ag cooperative MFA and their rancher members already participating in its Health Track program, as they provided a large data set of active cow-calf rancher users in BlockTrust's network (MFA Health Track, 2023).

Colorado State University's AgNext program was then added as a project partner to incorporate the USRSB's self-reported sustainability assessments, and introduce a methodology to aggregate these assessments so that individual cattle would have a full value chain sustainability score. So as cattle traveled from cow-calf ranch, to auction house, to feed yard, it would pick up unique sustainability scores associated with each step of the value chain that was traced via BlockTrust.

Data Analysis

Sustainability Scoring Methodology

In order to measure beef sustainability performance, this project sought to utilize and build upon industry and market accepted methodology, the best of which was determined to be the USRSB's U.S. Beef Industry Sustainability Framework (USRSB Framework, 2023). The framework is a resource developed by the USRSB to identify opportunities for continuous improvement, establish benchmarks for current performance, and assess progress toward goals critical to the U.S. beef industry's sustainability efforts. The Framework is comprised of:

- *High priority indicators (6): animal health and well-being, efficiency and yield, water resources, land resources, air and greenhouse gas emissions, and employee safety and well-being. These indicators are the foundation of the Framework, defining the areas most important to beef sustainability for the entire supply chain.*

- *Metrics: Metrics measure activities linked to each of the high priority indicators. The metrics were developed to address the unique characteristics of each segment of the value chain, outlining ways an operation or company can measure sustainability progress. The approach and development of metrics was owned by each value chain sector with an expectation to actively engage other stakeholder groups including civil society and allied industry members.*

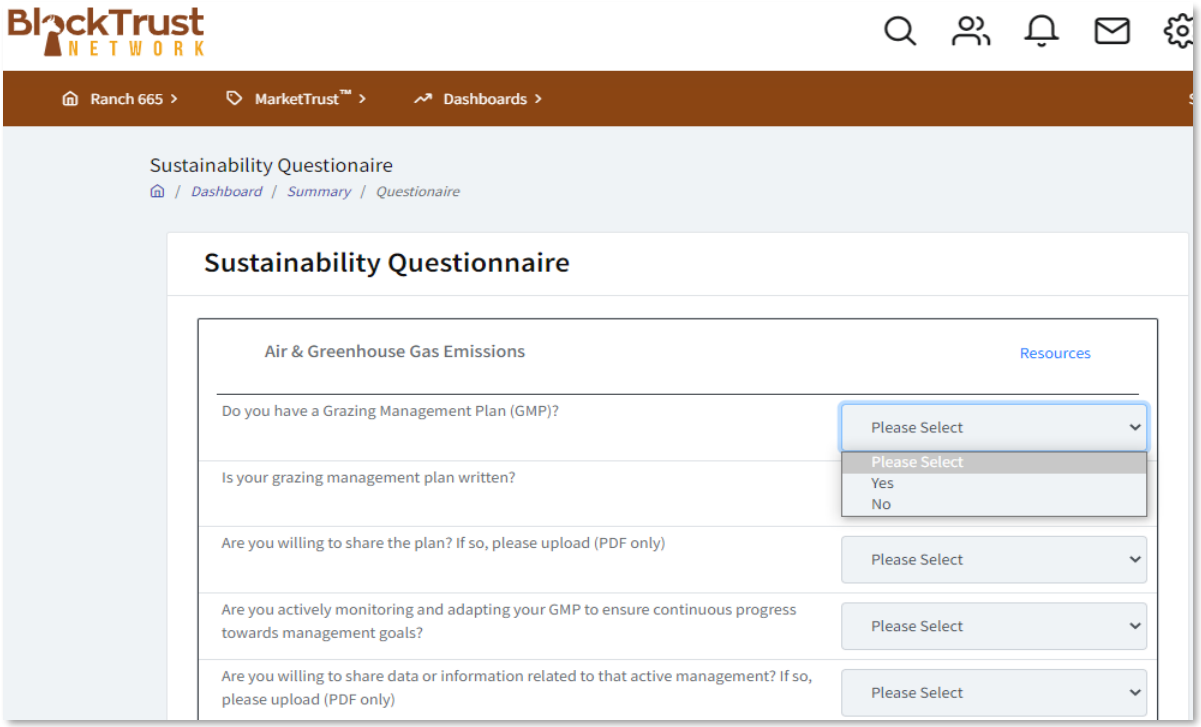
- *Sustainability (Self)-Assessment Guides: To operationalize the metrics, a technical guidance document was needed to aid user understanding and facilitate implementation. This effort led to development of the Sustainability Assessment Guides (SAGs) for each sector which provide value chain members additional tools and resources for assessing their own operation in relation to the high priority indicators and accompanying metrics. Much like the metric development process, value chain sectors took the lead in developing these resources utilizing feedback received from every sector. The SAGs outline the purpose and resources for approaches and methods to improve each metric.*

CSU AgNext created a sustainability scorecard that applied the six USRSB high-priority indicators, metrics, and targets emphasized for each sector. A scorecard was developed for each of the beef sectors, which includes:

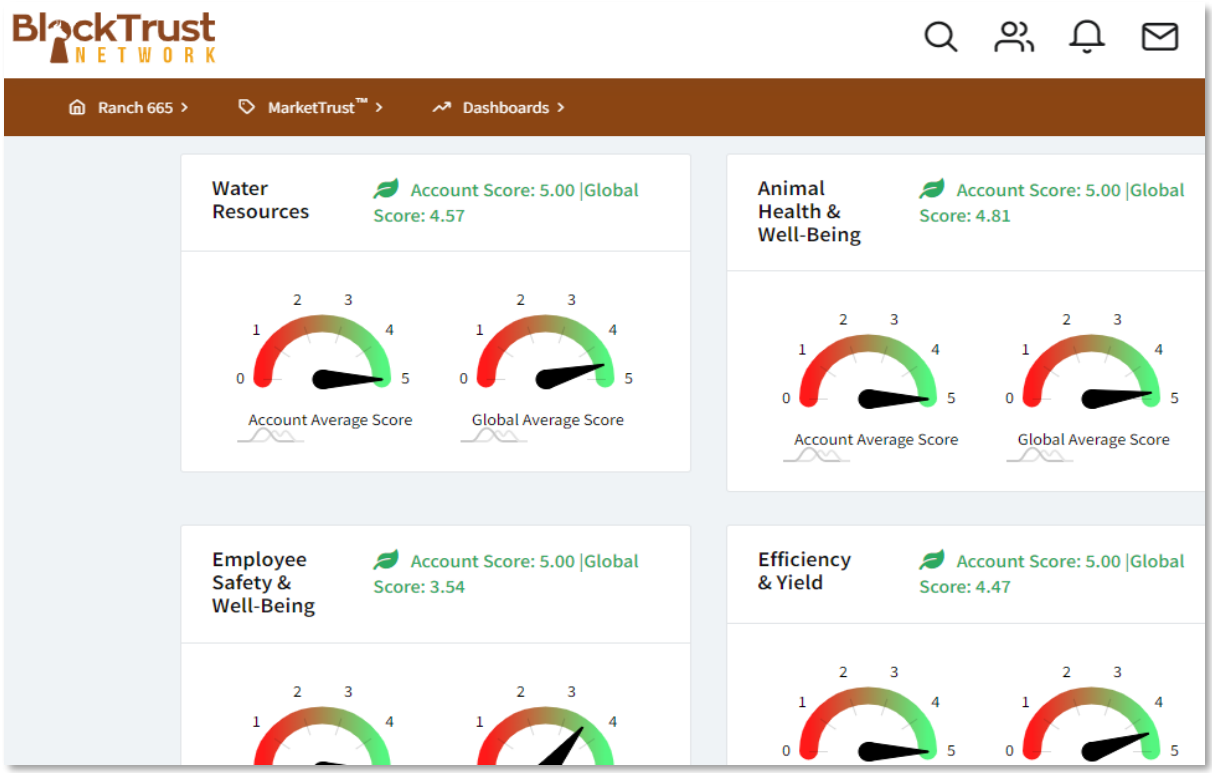
1) Cow-calf (also represents the stocker and backgrounder operations),
2) Livestock auction market
3) Feedyards
4) Packing & processing plants
5) Retail & food service.

All scorecards are worth 50 points and each high-priority indicator is worth 5 points, no matter the sector. The total sustainability score was calculated by averaging the scores of each high-priority indicator and generating a percentage. An additional extra credit section was developed by Rabo AgriFinance to reinforce the USRSB high-priority indicators, metrics, and targets to encourage continuous improvement regarding sustainability-related topics. Through the use of blockchain technology, the percentage score is tallied along the value chain to produce a total sustainability score of all the sectors combined and reported as a percentage. AgNext and Rabo AgriFinance recognize that some companies may choose to report at a facility, business unit, or company level. The scorecards were designed within each sector with this in mind, so the scores were not discounted for either approach.

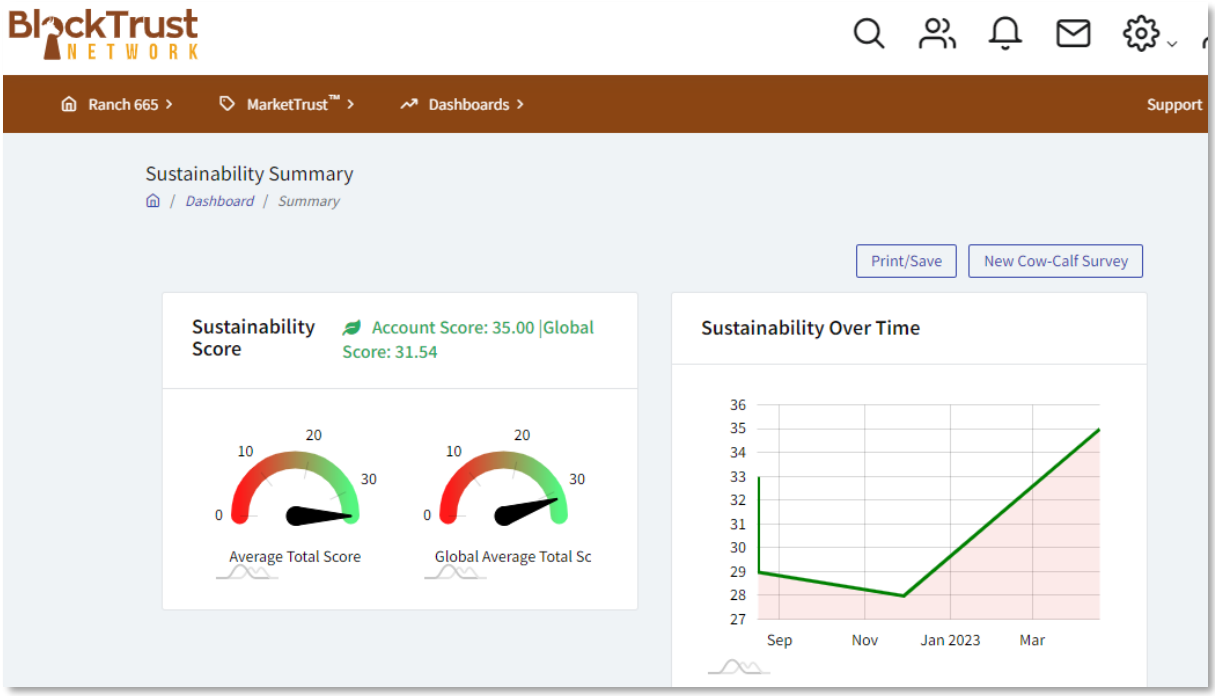
Once completed, this sustainability scoring methodology was incorporated within the BlockTrust platform for users to self-report their sustainability performance as an attribute traced with individual cattle (see images below).



BlockTrust Screenshot: Cow-Calf Rancher view when completing the Sustainability Questionnaire, 2023



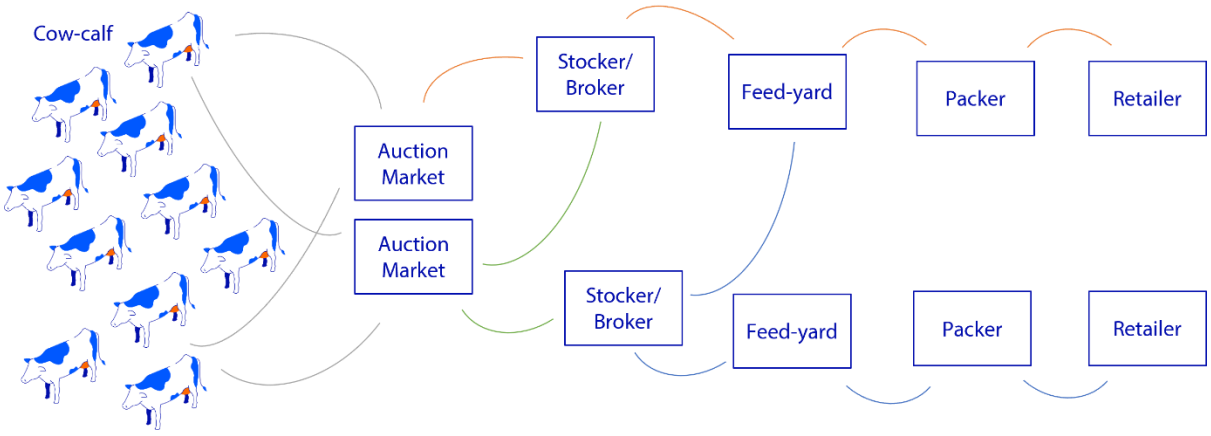
BlockTrust Screenshot: Cow-Calf Rancher view of their Sustainability Score by High Priority Indicator, 2023



BlockTrust Screenshot: Cow-Calf Rancher view of their Sustainability Score Compared to Global Average, 2023

While the above methodology produced a Sustainability Performance Score that allows the project to provide insights into the sustainability of the beef supply chain, it does have its flaws as the information is self-reported and subjective. In future phases, the focus will be to transition to more objective and valuable sustainability data, specifically carbon footprint information.

After the cattle were traced from cow-calf ranch to feed yard, beef processors were added to the data chain in 2022 to provide two unique data sets for analysis in 2023:



Live Cattle Data Set:

The principal intention of this beef traceability project was to trace live cattle through the beef supply chain and assign sustainability performance scores based on their journey, and this goal was accomplished. Live cattle were tracked from a number of MFA cow-calf ranches in 2021-2022, to feedlot, and all the way to the beef processing plant in the summer of 2023, resulting in a dataset of 314 unique cattle and sustainability scores.

Sector	Account	Company	No. Tags	Sustainability Score	Supply Chain Sustainability Score
Cow-Calf	728	Ranch 728	58	16	25.67
	740	Ranch 740	56	20	27
	766	Ranch 766	24	22	27.67
	772	Ranch 772	8	13	24.67
	908	Ranch 908	4	15	25.33
	909	Ranch 909	64	24	28.33
	911	Ranch 911	5	24	28.33
	912	Ranch 912	28	5	22
	913	Ranch 913	40	23	28
	914	Ranch 914	25	20	27
	Weighted Average		314	**18.2**	
Feedlot	776	Feedlot 776	314	**28**	
Packer	1053	Packer 1053	314	**33**	
Total Value Chain Score	**26.4**				

Source: BlockTrust Dataset, 2023

This dataset successfully demonstrates that BlockTrust can trace cattle from cow-calf to packer, collecting sustainability performance information from each step of the supply chain and creates the ultimate sustainability score.

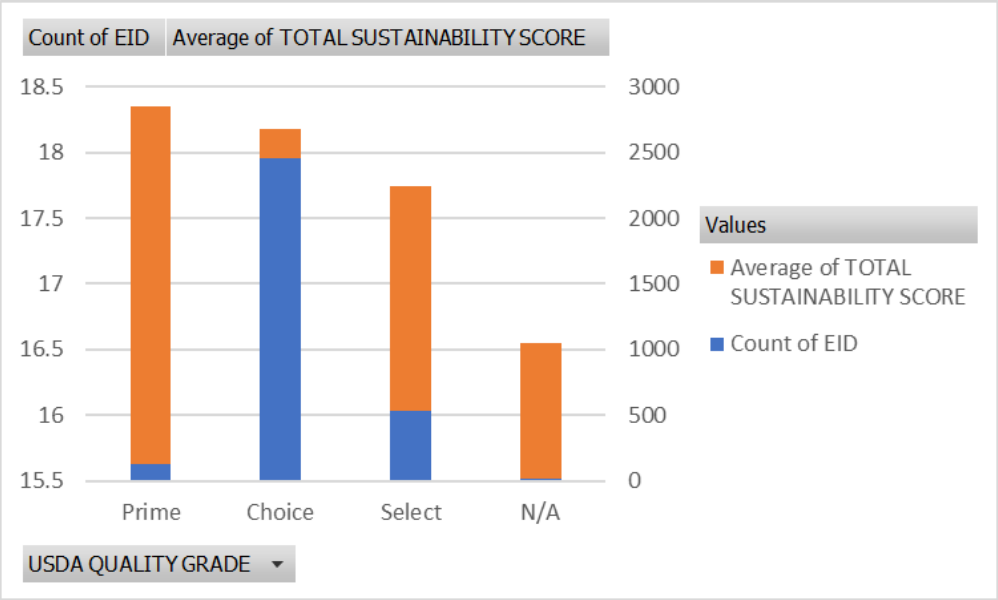
Historic Data Set:

For another packer, cattle ID tags in the BlockTrust database from cow-calf to feed yard were matched to processing plant data, identifying ~3,100 unique cattle traveling through the chain over the past 4 years. The sustainability scores from the cattle's unique journeys (~1,500) were accompanied by additional output data from the processing plant that included USDA Quality and Yield grades, which enabled initial analysis to explore the relationship between sustainability performance and cattle outcomes.

The USDA's Agricultural Marketing Service is the agency responsible for grading beef. Grading for quality means evaluation of traits related to tenderness, juiciness, and flavor (USDA, 2023).

- *USDA Quality Grade*
  - USDA-graded beef sold at the retail level is Prime, Choice, and Select, with Prime being most valuable and Select least of the three.
- *USDA Yield Grade*
  - The lower the numerical value of the USDA yield grade, the higher the expected yield of closely trimmed, boneless retail cuts. A yield grade 1 carcass provides the greatest amount of saleable beef while a yield grade 5 is the lowest-yielding carcass.

While most cattle in the data set graded at *Choice*, a positive correlation was observed between *Sustainability Score* and *USDA Quality Grade*.



Source: BlockTrust Dataset, 2023

Upon deeper data analysis (correlation and regression), the positive correlation between *Sustainability Score* and *USDA Quality Grade* or *USDA Yield Grade* were proven to not be

statistically significant. Which isn't to say that a positive correlation doesn't exist, but that the data set was too limited to answer this particular question.

If there were a statically significant, positive correlation between *Sustainability Score* and *USDA Quality Grade,* there would exist the potential for packers to see greater economic yield outcomes associated with sourcing more sustainable cattle. Beyond the motivation to source more sustainable cattle in order to make progress against climate goals, the fact that those cattle receive higher USDA Quality Grade means that their end product would be more valuable.

Exploring Business Models for Sustainably Traced Beef

While this project sought to build a platform that could trace cattle and their sustainability across the US beef supply chain, in 2023 it also began exploring business models that would support a business case for more sustainable beef. Rabobank's Research and Value Chain Finance teams were engaged in structured brainstorms, and on an ongoing basis, to explore this topic in detail. Ultimately both Supplier Finance and Structured Inventory Products (SIP) were seen as potential solutions. The initial findings occurred before the above data sets became available, which will make a continuation of this exercise key to the next steps of this project.

A primary challenge for traced, sustainable beef is ensuring that any premium or price signal provided by retail can make its way through the supply chain and to the cow-calf producers in meaningful and incentivizing amount. The project started by looking at the intersection of feed lots and processing plants where pricing is determined by a Formula/Grid Component.

Quality/Program	2022 Avg.	% Base Price
USDA Prime	$20	9%
USDA Choice	PAR	PAR
USDA Select	-$18	-8%
USDA Standard	-$32	-14%
Certified Angus Beef	$5	2%
All Natural	$33	15%
Non-Hormone Treated Cattle	$22	10%
>30 Months of Age	-$20	-9%

The 2022 average formula dressed steer base price was $226/cwt.

The initial research led to a number of questions to pursue moving forward: Could a traceability component be added to a standard fed cattle pricing formula? Age-and-Source Verification created a market-place premium briefly after BSE was discovered in the United States, but formal grids only presented a discount for cattle over 30 months of age. How could a traceability grid component be structured to get the necessary price signals back to the cow-calf segment?

Key Learnings

- The Beef Traceability Project, with BlockTrust's technology, can trace cattle from cow-calf to packer, collecting sustainability performance information from each step to provide a supply chain Sustainability Performance Score
- Additional data and analysis may demonstrate a statistically significant, positive correlation between *Sustainability Score* and *USDA Quality Grade,* which would support the opportunity for packers to see greater economic yield outcomes associated with sourcing more sustainable cattle
- The business case for traced, more sustainable cattle will require a coordinated effort across the entire value chain to ensure that any price signals ultimately reaches the cow-calf ranchers

Next Steps

The project to date has built a powerful infrastructure of partners, data, and learnings to build on moving forward. Moving forward, this project should address the following:

1) Improve sustainability data

a. The aggregated USRSB self-reported sustainability scores represented a great "proxy" for sustainability performance, but are flawed in their "self-reported" nature and lack a key data point of interest: carbon footprint
b. Add beef carbon footprint data
    i. Rabobank and its Rabo Trace team have engaged a data and analytics provider offering agriculture and sustainability insights, as a new partner to add a carbon data layer, more objective in nature and directly useful for beef companies looking to better understand the carbon footprint of their supply chains
    ii. Rabobank is also working with AgNext to develop a user(farmer)-friendly carbon calculator that could also be utilized in parallel
c. Explore genetic data
    i. Genetic selection has long included traits on growth, carcass, and reproduction, but recently more focus has been placed on traits that impact efficiency and sustainability such as mature cow weight, maintenance energy, and methane emissions
    ii. Might there be a business case for beef companies to source cattle with low impact genetics?

2) Identify business models for more traced, more sustainable, and/or lower carbon beef

a. Further explore Value Chain Finance products and mechanisms that might incentivize more sustainable cattle through their business models.
b. Can traceability and ultimately carbon footprint be added to the standard fed cattle pricing grid?
c. Can we eliminate market inefficiencies and production gaps at the cow calf level for value added beef?

3) Increase Cow-Calf rancher representation

a. While the US beef processing sector is quite concentrated in four players, the cow-calf rancher sector is very diverse and dispersed. The project should look to add a variety of cow-calf representing organizations to increase participation and the cow-calf data set. Cow-calf ranchers, especially those with smaller herds, can be difficult to engage in such projects.
    i. How much value would they see in receiving their cattle's yield and quality grades back?
    ii. How much more would they need to be paid to demonstrate more sustainable cattle?
b. One opportunity might be to align with the ongoing work of the $30 million USDA Climate Smart Commodity Grant led by the American Farmland Trust

4) Increase sector coverage and complete the value chains

a. While 2/4 of the major packers are in the project represented, the last two should be engaged to create a US industry approach to beef traceability and sustainability
b. Continue down the chain to interested retail stakeholders to explore the business case and price signaling for traced, more sustainable beef

Special thanks to the Project's Partners and all those who invest their time, energy, and resources in the pursuit of understanding and addressing the climate impacts of the US Beef sector.

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